LAB 7

Animation

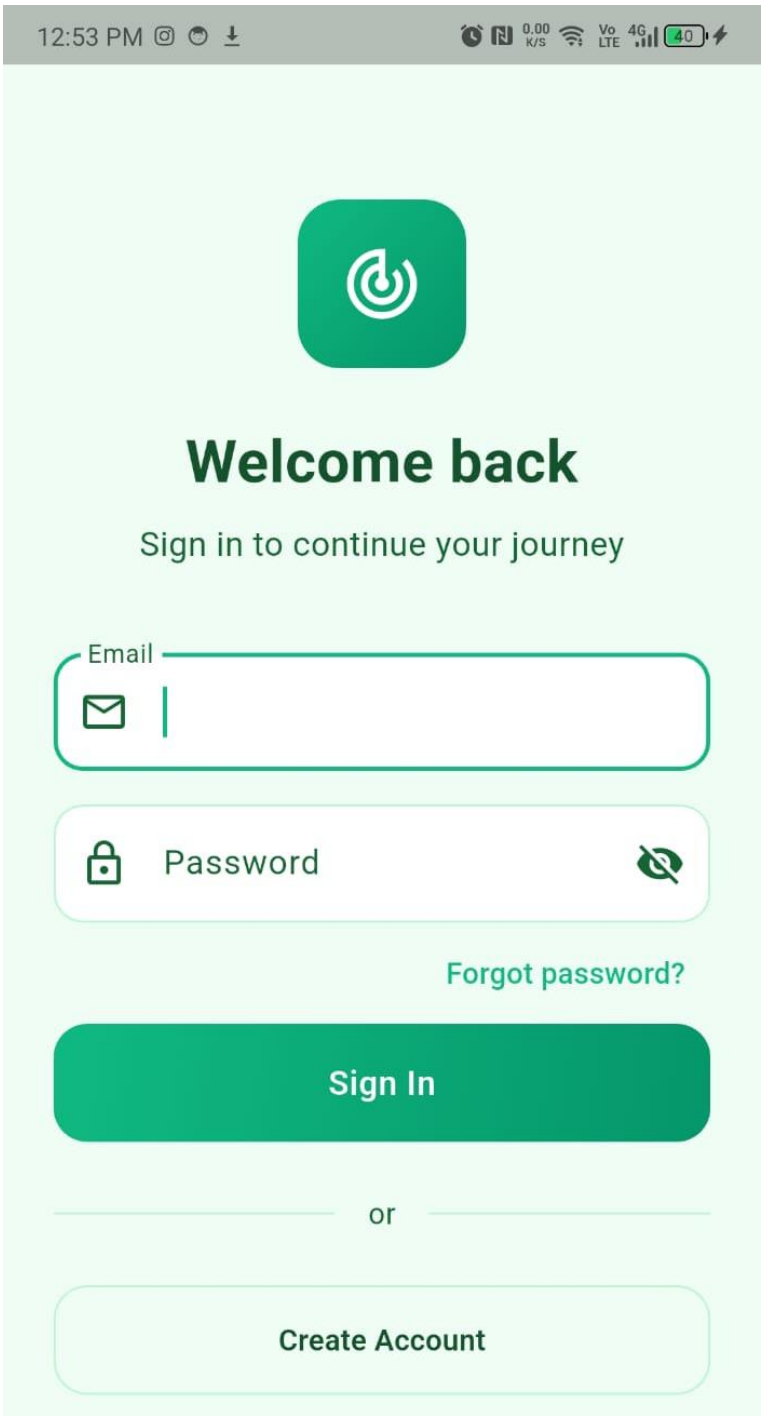
ANIMATION:

Slide Transition + Fade Transition -> Entrance Animation

```
late AnimationController _entranceCtrl;
late Animation<Offset> _slideAnim;
late Animation<double> _fadeAnim;

@override
void initState() {
  super.initState();
  _entranceCtrl = AnimationController(
    vsync: this,
    duration: const Duration(milliseconds: 700),
  );
  _slideAnim = Tween<Offset>(
    begin: const Offset(0, 0.35),
    end: Offset.zero,
  ).animate(CurvedAnimation(parent: _entranceCtrl, curve: Curves.easeOutCubic));
  _fadeAnim = Tween<double>(begin: 0, end: 1)
      .animate(CurvedAnimation(parent: _entranceCtrl, curve: Curves.easeIn));
  _entranceCtrl.forward();
}
```

```
return Scaffold(
  backgroundColor: theme.background,
  body: SafeArea(
    child: FadeTransition(
      opacity: _fadeAnim,
      child: SlideTransition(
        position: _slideAnim,
        child: SingleChildScrollView(
      padding: const EdgeInsets.all(24),
```



Login Screen Animation:

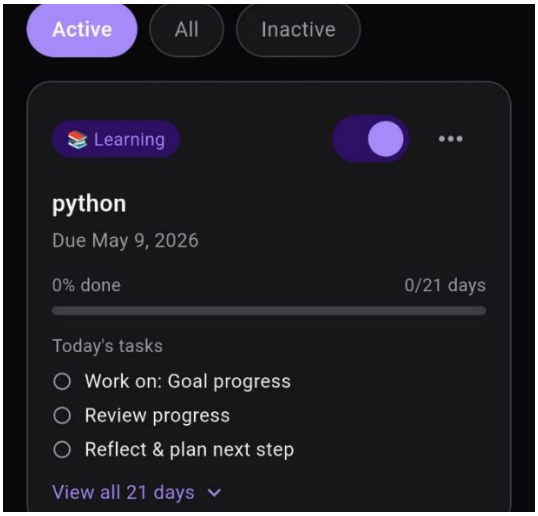
- Animation can’t be properly seen within screenshots
- Every time the Login screen opens
- the logo, title, input fields and buttons all slide up smoothly
- from 35% below while fading from invisible to fully visible
- takes 700ms with easeOutCubic curve for a natural feel.

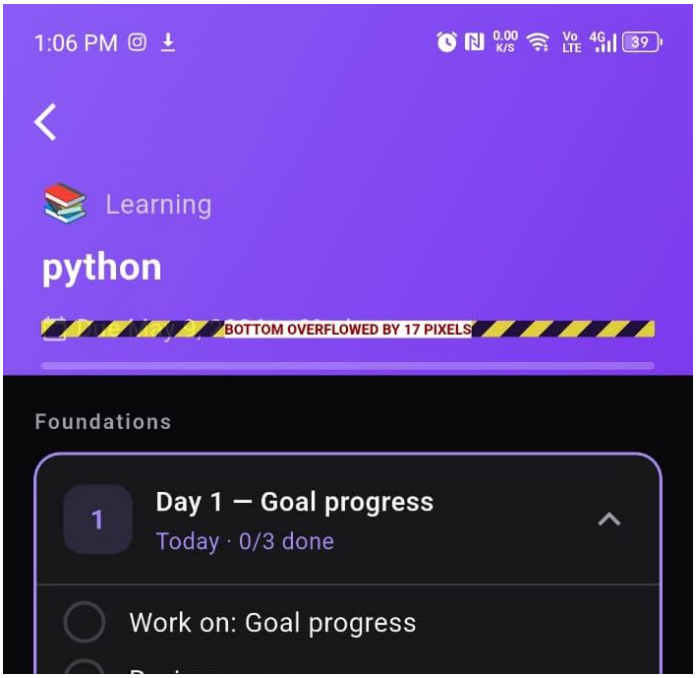
Goal Screen :

```
child: Hero(
  tag: 'goal-name-${goal.id}',
  child: Material(
    color: Colors.transparent,
    child: Text(
      goal.name,
      style: TextStyle(
        color: theme.text,
        fontSize: 16,
        fontWeight: FontWeight.w600,
      ),
    ),
  ),
```

Goal details screen:

```
Hero(
  tag: 'goal-name-${goal.id}',
  child: Material(
    color: Colors.transparent,
    child: Text(
      goal.name,
      style: const TextStyle(
        color: Colors.white,
        fontSize: 20,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
),
```



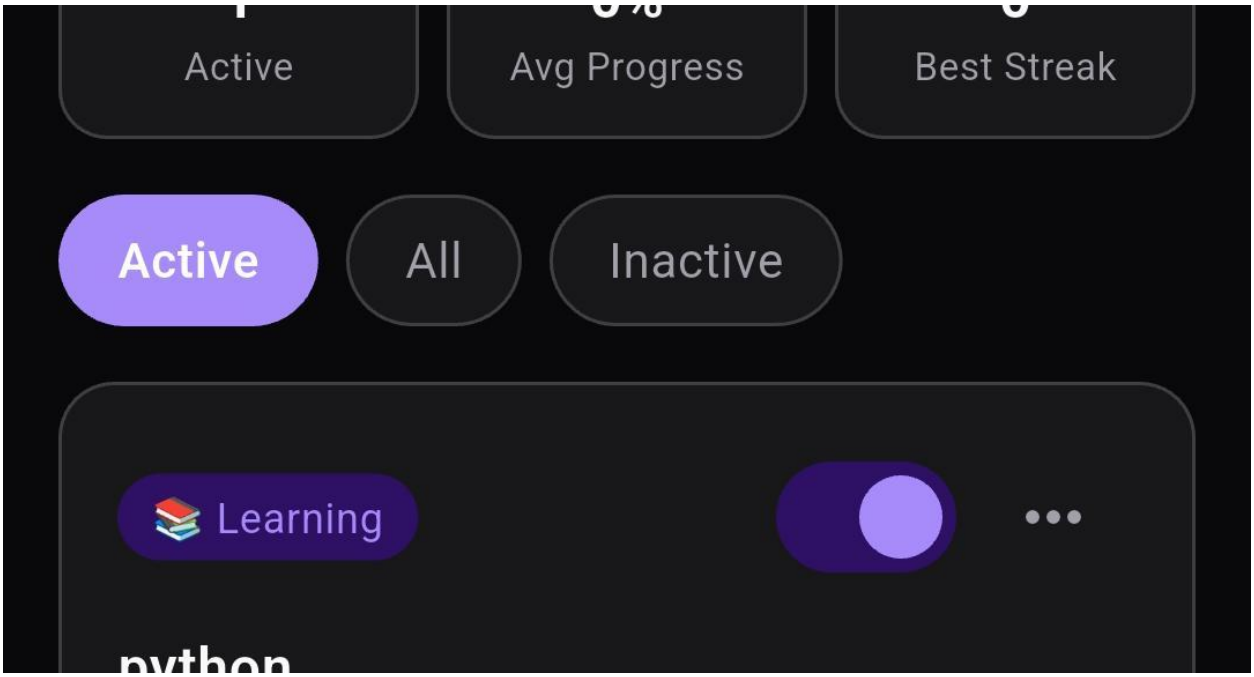


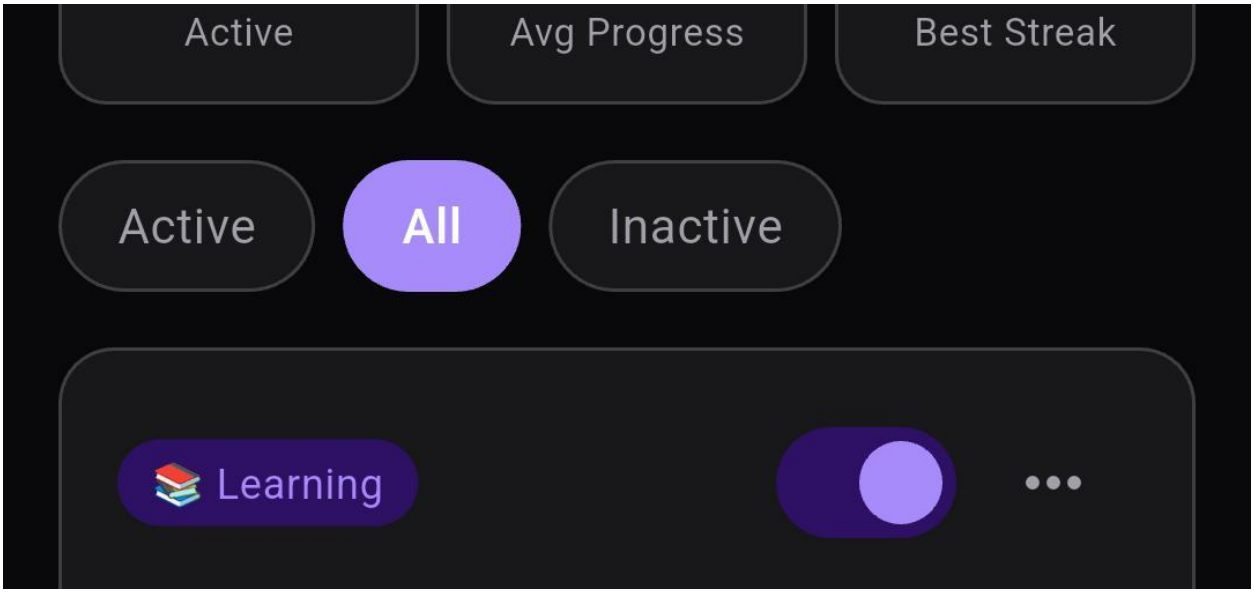
Sliding upward animation:

- Animation can’t be properly seen within screenshots
- Tap any goal card in the Goals tab
- Goal's title text physically flies from the card into the detail screen
- Header (white, 20px), morphing size and color mid-flight.
- Flutter handles the transition automatically using the matching tag.

Goal Screen

```
176          else
177            SliverToBoxAdapter(
178              child: AnimatedSwitcher(
179                duration: const Duration(milliseconds: 350),
180                switchInCurve: Curves.easeOut,
181                switchOutCurve: Curves.easeIn,
182                transitionBuilder: (child, animation) => FadeTransition(
183                  opacity: animation,
184                  child: SlideTransition(
185                    position: Tween<Offset>(
186                      begin: const Offset(0, 0.08),
187                      end: Offset.zero,
188                    ).animate(animation),
189                    child: child,
190                  ),
191                ),
192                child: Column(
193                  key: ValueKey(_filter),
194                  children: List.generate(
195                    filtered.length,
196                    (index) => Padding(
197                      padding: const EdgeInsets.fromLTRB(16, 0, 16, 12),
198                      child: _GoalCard(goal: filtered[index], theme: theme),
199                    ),
200                  ),
201                ),
```





Goal Pill Animation:

- Animation can't be properly seen within screenshots
- In **Goals tab**, the filter pills say **Active / All / Inactive**.
- Tapping a different filter causes the old list to **fade out + slide away** and the new list to **fade in + slide up** smooth 350ms crossfade with slide effect.